

# Rifqi Afta Ronanda

IT and AI Engineering

## Turning data into useful decisions.

I connect engineering thinking, machine learning, and clear interfaces to build practical tools for research and operations.

Engineering Physics background at Universitas Gadjah Mada. Exploring IT and AI engineering opportunities with companies in Japan.

### FOCUS

Practical data workflows, machine learning, responsive web applications, and clear technical communication.

### SELECTED WORK

Solar operations monitor | Local AI evidence database | Quantum banking research hub | Soundscape analysis pipeline | Manabi Japanese learning dashboard

### HOW TO READ THIS PORTFOLIO

Each case separates the implemented workflow from its intended business value and remaining validation. Commercial outcomes, production deployments, and performance gains are not claimed without evidence.

GitHub / [rifqironanda](#)

LinkedIn / [Rifqi Afta Ronanda](#)

Repository documentation reviewed: 7 October 2026. LinkedIn is provided as a profile link; detailed employment information has not been independently reviewed for this edition.

## 01 / Solar operations monitor

Research prototype | Python / Streamlit / Isolation Forest

### PROBLEM

Low solar output can come from weather or unusual operating conditions. Teams need context before deciding what to inspect.

### IMPLEMENTATION

A CSV-to-dashboard workflow combining data validation, an irradiance and temperature baseline, Isolation Forest per inverter, and explainable alerts.

### INTENDED BUSINESS VALUE

Designed to help operations teams prioritise inspections and interpret underperformance with environmental context.

### EVIDENCE

Repository documents CSV validation, inverter filtering, actual-versus-expected charts, and alert-logic unit tests.

### VALIDATION NEEDED

No measured downtime reduction or verified fault-detection accuracy. Ambient temperature is a proxy; plant-specific calibration and maintenance records are needed.

[View project documentation on GitHub](#)

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## 02 / Local AI evidence database

Local MVP | Python / Ollama / SQLite

### PROBLEM

Literature review involves repetitive PDF extraction, summarisation, and organising evidence for later review.

### IMPLEMENTATION

Local PDF discovery and page-text extraction with PyMuPDF, Ollama summaries, SQLite storage, and a manual validation state.

### INTENDED BUSINESS VALUE

Designed to reduce repetitive document handling while keeping processing local and human review explicit.

### EVIDENCE

Documented architecture separates documents, page text, summaries, and validations. No cloud LLM API is required by the default pipeline.

### VALIDATION NEEDED

No measured time savings. Reliable claim extraction and citation verification remain roadmap items; summaries require human validation.

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## 03 / Quantum banking research hub

Interactive web application | React / Supabase / CSS

### PROBLEM

Research materials, demonstrations, and report drafts can be difficult to navigate when they live in separate tools.

### IMPLEMENTATION

A React dashboard with 30 presentation materials, interactive labs, an outline editor, local-first storage, and optional Supabase synchronisation.

### INTENDED BUSINESS VALUE

Brings reading, demonstrations, and drafting into one workflow to support research communication and structured review.

### EVIDENCE

Repository documents responsive navigation, an outline editor, Supabase Auth and Row Level Security, revision history, and offline fallback.

### VALIDATION NEEDED

Research communication tool. No claim of bank deployment, regulatory endorsement, or measured productivity improvement.

[View project documentation on GitHub](#)

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## 04 / Soundscape analysis pipeline

Research pipeline | Python / MFCC / PCA / KMeans

### PROBLEM

Audio collections need consistent representations before similarities and unusual recordings can be explored.

### IMPLEMENTATION

MFCC, spectral and temporal features; statistical aggregation; PCA; KMeans clustering; and Isolation Forest outlier detection.

### INTENDED BUSINESS VALUE

Creates reusable features, cluster labels, and outlier indicators to support exploratory review of environmental recordings.

### EVIDENCE

Repository specifies feature, label, and outlier array outputs with a repeatable Python workflow.

### VALIDATION NEEDED

Clusters are not verified species or semantic labels. No demonstrated biodiversity metric or commercial monitoring outcome.

[View project documentation on GitHub](#)

## 05 / Manabi Japanese learning dashboard

Learning application | React / Web Speech API / localStorage

### PROBLEM

Self-study benefits from a clear path through lessons, practice, feedback, and progress tracking.

### IMPLEMENTATION

A responsive React interface with 18 Irodori Starter lessons, text-to-speech, graded quizzes, timed practice, and local progress storage.

### INTENDED BUSINESS VALUE

Supports independent study through immediate explanations and a consistent lesson-to-practice workflow.

### EVIDENCE

Repository documents 18 lessons, 30-question practice sets, responsive layouts, and localStorage progress.

### VALIDATION NEEDED

Independent practice application, not an official JLPT or JFT-Basic simulator. Learning gains and language proficiency are not claimed.

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